

Calculus 1M, 2019, GTIIT,

Midterm exam, November 18, 2019

November 11, 2019

1 Multiple choice (8 points for each question)

Denote by "V" your final choice in the appropriate spot of the table on the second page of this questionnaire.

1. Consider the function $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x|x|$, for $x \in \mathbb{R}$. Then, (choose the correct statement from the following options)
 - (a) f is not differentiable at $x = 0$.
 - (b) $f'(0) = 1$ and $f'(x)$ is continuous at $x = 0$.
 - (c) $f'(0) = 0$ and $f'(x)$ is not continuous at $x = 0$.
 - (d) $f'(0) = 0$ **and** $f'(x)$ **is continuous at** $x = 0$.
 - (e) $f'(0) = 1$ and $f'(x)$ is not continuous at $x = 0$.

2. Consider the function $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = \frac{1}{1+|x|} + \frac{1}{1+|x-2|}$, for $x \in \mathbb{R}$ and let $Im(f) = f(\mathbb{R})$. Then, (choose the correct statement from the following options)
- (a) $Im(f)$ is not bounded.
 - (b) $Im(f)$ has minimum which is strictly less than 1.
 - (c) $Im(f)$ has maximum which is strictly less than 1.
 - (d) $Im(f)$ has supremum but no maximum.
 - (e) $Im(f)$ **has infimum but no minimum.**
3. Let $f : [0, 1] \rightarrow [0, 1]$ be a continuous function. Then necessarily (choose the correct statement from the following options)
- (a) $\exists c \in [0, 1]$ such that $f(c) = \ln(\frac{\pi c}{2})$.
 - (b) $\nexists c \in [0, 1]$ such that $f(c) = \sin(\frac{\pi c}{2})$.
 - (c) $\exists c \in [0, 1]$ **such that** $f(c) = \cos(\frac{\pi c}{2})$.
 - (d) $\exists c \in [0, 1]$ such that $f(c) = \tan(\frac{\pi c}{8})$.
 - (e) $\nexists c \in [0, 1]$ such that $f(c) = \tan(\frac{\pi c}{8})$.
4. Choose the correct statement from the following options.
- (a) If $A \subset \mathbb{R}$ is not open, then A is closed.
 - (b) Every set $S \subset \mathbb{R}$ is either open or closed.
 - (c) The point $\{0\}$ is a limit point of every non-empty set $A \subset \mathbb{R}$ with $\inf A = 0$.
 - (d) ***There are open coverings of the interval $(0, 1)$ such that they have a finite subset which is a covering of $(0, 1)$.***
 - (e) Every non-empty bounded open set $S \subset \mathbb{R}$ has a largest element.

5. Choose the correct statement from the following options.

- (a) Let $B(x, r) = \{y \in \mathbb{R} \mid |x - y| < r\}$ for $x \in \mathbb{R}, r > 0$. The open covering $\mathcal{H} = \{B(z, r) \mid -1 < z < 1, r \in \mathbb{Q} \cap (0, 1)\}$ of the set $[0, 1)$ has a finite subcovering of $[0, 1)$ and hence $[0, 1)$ is compact.
- (b) If every point of a nonempty bounded set $A \subset \mathbb{R}$ is a limit point of A , then A is closed.
- (c) The set $V = \cup_{n=1}^{\infty} \{\frac{1}{n}\}$ is bounded below and each of the sets $\{\frac{1}{n}\}$ is closed. Therefore V has a minimum.
- (d) The boundary of a non-empty set $A \subset \mathbb{R}$ consists of those points z of A for which every punctured neighborhood of z contains at least one point not in A .
- (e) **The boundary of a non-empty set $A \subset \mathbb{R}$ can contain interior points.**

6. Choose the correct statement from the following options.

- (a) $f : \mathbb{R} \rightarrow \mathbb{R}$ is continuous everywhere if and only if $\forall \delta > 0 \exists \epsilon > 0$ such that $|x - y| < \delta \Rightarrow |f(x) - f(y)| < \epsilon$.
- (b) $f : \mathbb{R} \rightarrow \mathbb{R}$ is continuous everywhere if and only if $\forall \epsilon > 0 \exists \delta > 0$ such that $|x - y| < \delta \Rightarrow |f(x) - f(y)| < \epsilon$.
- (c) **Let $x_0 = 1, x_{k+1} = 1 - \frac{x_k}{3}, k = 0, 1, 2, \dots$. Then $|x_{k+1} - \frac{3}{4}| < 3^{-k}$ for all $k = 0, 1, 2, \dots$**
- (d) Let $f(x) = [x]$ be the integer part of x . Then

$$\sup_{0 < x < 1} f(x) = 1 \quad \text{and} \quad \inf_{0 < x < 1} f(x) = 0.$$

- (e) Because $\lim_{x \rightarrow 0} x \sin(1/x) = 0$ we also have $\lim_{x \rightarrow 0} \frac{1}{x} \sin(x) = 0$.

7. Choose the correct statement from the following options.

Consider the function $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = e^x \sin(x)$, $x \in \mathbb{R}$. Then

- (a) f has a maximum but not a minimum over the interval $(-\infty, 0]$.
- (b) f has a minimum but not a maximum over the interval $(-\infty, 0]$.
- (c) f **has a maximum and a minimum over the interval** $(-\infty, 0]$.
- (d) f is bounded over $(-\infty, 0]$ but it doesn't have a maximum and a minimum over the interval $(-\infty, 0]$.
- (e) f is not bounded over $(-\infty, 0]$.

8. Consider the function $f_p : \mathbb{R} \rightarrow \mathbb{R}$, $p > 0$, defined by

$$f_p(x) = \begin{cases} x^p \sin(1/x), & x \neq 0 \\ 0, & x = 0 \end{cases}$$

Choose the correct statement from the following options.

- (a) For all $p > 0 : \exists f'_p(0)$.
- (b) For $p = 1 : f_p(x)$ is not continuous at 0.
- (c) For $p = 2 : f'_p(0)$ exists and $f'_p(x)$ is continuous at 0.
- (d) **For** $p = 2 : f'_p(0)$ **exists and** $f'_p(x)$ **is not continuous at 0.**
- (e) For all $p \geq 1 : f'_p(0) = 0$ and $f'_p(x)$ is not continuous at 0.

2 True/False (4 points for each question)

In the following questions you need to fill the appropriate empty square which is located to the right of each question.

If you think that statement is False, write the letter "F" inside the square. If you think that the statement is true, write the letter "T" inside the square.

Please note that in this particular exercise you don't need to prove true statements nor to provide counter examples for false ones, you only need to decide whether they are true or false.

- (a) If $\lim_{x \rightarrow a} f(x) = L$ and $L > 0$ (L is a finite real number) then there exists a neighborhood of a such that $f(x) \geq 0$ for any x in that neighborhood. ☐ F
- (b) If $\lim_{x \rightarrow \infty} f(x) = L$ (L is a finite real number) then $\lim_{x \rightarrow \infty} f'(x) = 0$. ☐ F
- (c) Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$ be continuous and suppose that the limit $\lim_{x \rightarrow \infty} g(f(x))$ exists and is finite. Then also the limit $\lim_{x \rightarrow \infty} f(x)$ exists and is finite. ☐ F
- (d) If $f : [0, \infty) \rightarrow [0, \infty)$, is a bijection, then also $g : [0, \infty) \rightarrow [0, \infty)$, $g(x) = f(x^2)$ is a bijection. ☐ T
- (e) Every non-empty bounded and closed set has at least one limit point. ☐ F

3 Open questions (7 points for each question)

In this section you need to write your final proof in the appropriate place allocated for it under the word "Answer". If you need some draft, please ask from the monitor.

Question

Prove using $\varepsilon - \delta$ definition.

$$\lim_{x \rightarrow 2} x^3 = 8.$$

Answer

Let $\varepsilon > 0$. We need to find $\delta > 0$ s.t. $0 < |x - 2| < \delta \implies |x^3 - 8| < \varepsilon$.

Draft

$$\begin{aligned} |x^3 - 8| &= |x-2||x^2+2x+4| \stackrel{\Delta\text{-ineq.}}{\leq} |x-2|(|x^2+2x|+4) = 4|x-2|+|x-2||x||x+2| \stackrel{(*)}{\stackrel{|x-2|<1}{<}} 4|x-2|+3 \cdot 5 \cdot |x-2| \\ &= 19|x-2| \stackrel{|x-2|<\frac{\varepsilon}{19}}{<} \varepsilon. \end{aligned}$$

Therefore, take $\delta = \min\{1, \frac{\varepsilon}{19}\}$.

(*)

$$|x - 2| < 1$$

$$-1 < x - 2 < 1$$

$$-3 < 1 < x < 3 \rightarrow |x| < 3$$

$$-5 < 3 < x + 2 < 5 \rightarrow |x + 2| < 5$$

End of Draft.

Back to proof:

Take $\delta = \min\{1, \frac{\varepsilon}{19}\}$. Then,

$$|x^3 - 8| \stackrel{\text{Draft}}{\leq} 4|x - 2| + |x - 2||x||x + 2| \stackrel{|x-2|<\delta \leq 1}{<} 19|x - 2| \stackrel{|x-2|<\delta \leq \frac{\varepsilon}{19}}{<} 19 \frac{\varepsilon}{19} = \varepsilon \quad \blacksquare.$$

Question

Let A and B be two nonempty sets of real numbers. Prove

$$\sup(A \cup B) = \max\{\sup(A), \sup(B)\}.$$

Answer

If $\sup(A) \geq \sup(B)$, then $\sup(A \cup B) = \sup(A)$.

If $\sup(B) \geq \sup(A)$, then $\sup(A \cup B) = \sup(B)$.

Hence, $\sup(A \cup B) = \max\{\sup(A), \sup(B)\}$ since the definition of the max function is satisfied. ■

Question

Prove using $\varepsilon - \delta$ definition that

if $\lim_{x \rightarrow a} f(x) = L$ then $\lim_{x \rightarrow a} |f(x)| = |L|$ where L is a finite real number.

Answer

Let $\varepsilon > 0$. We need to find $\delta > 0$ s.t. $0 < |x - a| < \delta \implies ||f(x)| - |L|| < \varepsilon$.

From the data we know that $\forall \varepsilon > 0, \exists \delta > 0$ s.t. $0 < |x - a| < \delta \implies |f(x) - L| < \varepsilon$.

For the fixed ε that we have started with, take the $\delta > 0$ assigned to it from the data above.

Then, we obtain

$$||f(x)| - |L|| \stackrel{\Delta\text{-ineq.}}{\leq} |f(x) - L| < \varepsilon$$

and we are done. ■

Notice that in this proof we used the triangle inequality in this version

$$||a| - |b|| \leq |a - b|, \quad \forall a, b \in \mathbb{R}.$$